

Trial Test 10

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1) Evaluate:

$$\frac{(3x^2-6x)(4x-3)}{(x-2)}$$

a) $12x^2 - 9x$

b) $14x^2 - 7x$

c) $16x^2 - 5x$

d) $18x^2 - 3x$

2) Given the sequence that $a_1 = 1$, $a_{n+1} - a_n = 3^n$.

Find the value of a_5 .

a) 81

b) 101

c) 121

d) 141

3) Solve

$$4x^4 + 9x^2 - 63 = 0$$

a) $x = \pm \sqrt{2}$

b) $x = \pm \sqrt{3}$

c) $x = \pm \sqrt{5}$

d) $x = \pm \sqrt{7}$

4) Find the area of parallelogram, if given that the diagonals are equal to 15 and 20 respectively, and the angle between them is equal to 60°

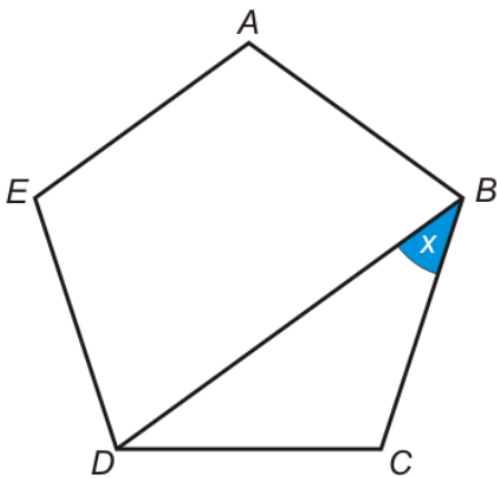
a) $55\sqrt{3}$

b) $65\sqrt{3}$

c) $75\sqrt{3}$

d) $85\sqrt{3}$

5) Find the value of x in the given regular polygon



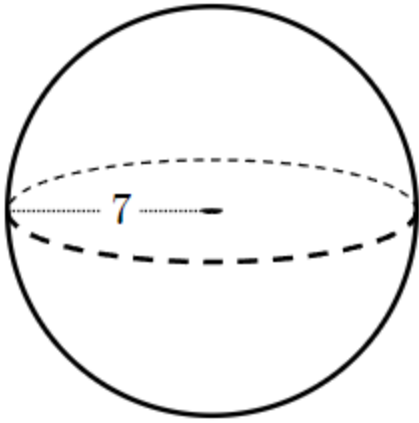
a) 24

b) 36

c) 48

d) 72

6) Find the volume of a given sphere



a) $\frac{49\pi}{3}$

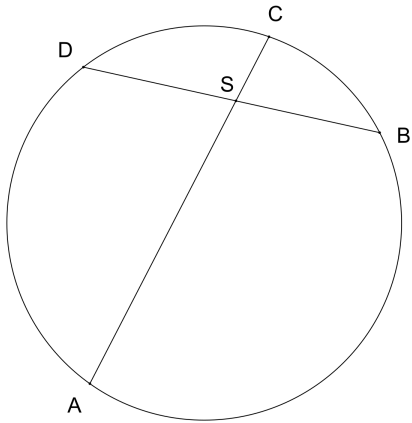
b) $\frac{343\pi}{3}$

c) $\frac{1029\pi}{4}$

d) $\frac{1372\pi}{3}$

7) Given that $DS = 6$, $SB = 8$, $AS = 12$.

Find the value of CS



- a) 1
- b) 2
- c) 3
- d) 4

8) Rearrange the formula in terms of Y:

$$X = \sqrt{19 - \sqrt{25 - 7(Y + 9)^2}}$$

$$\text{a) } Y = \sqrt{-\frac{(17-X^2)^2+25}{7}} - 9$$

$$\text{b) } Y = \sqrt{-\frac{(19-X^2)^2-25}{7}} - 9$$

$$\text{c) } R = \sqrt{\frac{(X^2+17)^2+25}{2}} + 9$$

$$d) R = \sqrt{\frac{(X-17)^2 - 25}{2}} + 9$$

9) Find the sum of infinite geometric sequence, if given that $b_2 = 15$ and $b_5 = \frac{5}{9}$

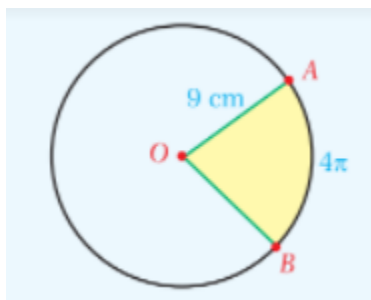
a) $\frac{125}{2}$

b) $\frac{135}{2}$

c) $\frac{145}{2}$

d) $\frac{155}{2}$

10) Find the area of shaded region



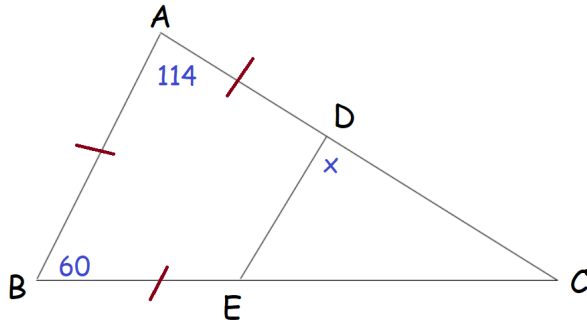
a) $14\pi \text{ cm}^2$

b) $16\pi \text{ cm}^2$

c) $18\pi \text{ cm}^2$

d) $22\pi \text{ cm}^2$

11) Find x



(figure is not to scale)

- a) 84
- b) 87
- c) 114
- d) 117

12) Find the surface area of the cone, if given that radius is equal to 5 cm and height is equal to 12 cm

- a) 90π
- b) 96π
- c) 112π
- d) 126π

13) When the expression $(2x - 2)^2 - (x + 8)^2$ written in the form $p(x + q)^2 + r$ where p, q and r are constants, what is the value of r ?

- a) 96
- b) 108

c) -96

d) -108

14) Convert 12697 liters to m^3

a) 1,2697 m^3

b) 12,697 m^3

c) 126,97 m^3

d) 1269,7 m^3

15) Evaluate and write in standard form

$$\frac{55 \cdot 10^{19}}{4 \cdot 10^5}$$

a) $137.5 \cdot 10^{13}$

b) $13.75 \cdot 10^{14}$

c) $1.375 \cdot 10^{15}$

d) $0.1375 \cdot 10^{16}$

16) A medical scanner is bought for \$15000. The value of the scanner depreciates by 20% every year.

By how much has the scanner reduced in value after 2 years?

a) \$5400

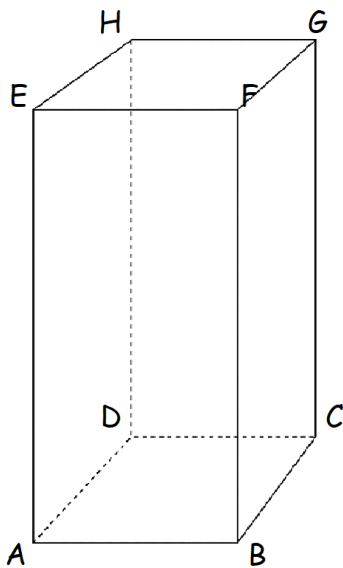
b) \$6000

c) \$9000

d) \$9000

17) In the rectangular solid in the adjacent figure, $AB = 3$, $BF = 12$, $BC = 4$.

Find the length of AG



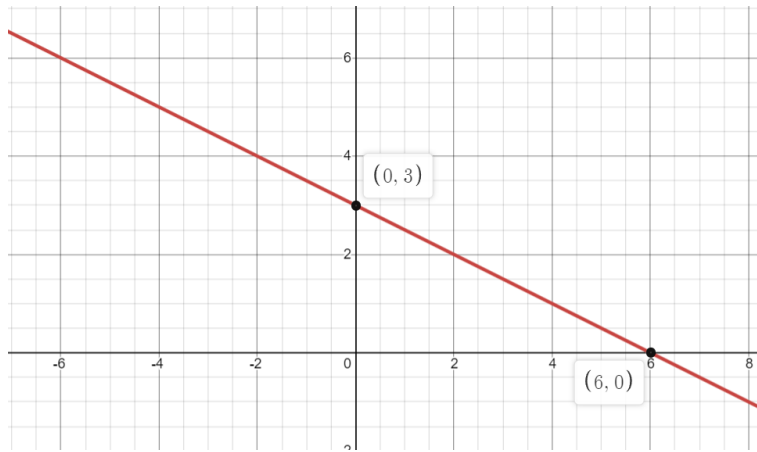
a) 11

b) 13

c) 17

d) 21

18) Find the equation of the line:



a) $y = -\frac{1}{2}x + 3$

b) $y = -\frac{1}{3}x + 6$

c) $y = -\frac{1}{4}x - 3$

d) $y = -\frac{1}{5}x - 6$

19) The quantities x and y are positive. x is inversely proportional to the square root of y .

When $x = 8$, $y = 9$. What is the value of y when $x = 6$

a) 12

b) 16

c) $\frac{81}{16}$

d) $\frac{27}{14}$

20) ABCD is a square. Sides AB and CD are each increased in length by 40%. Sides BC and AD are each decreased in length by 75%. ABCD has been transformed into a rectangle.

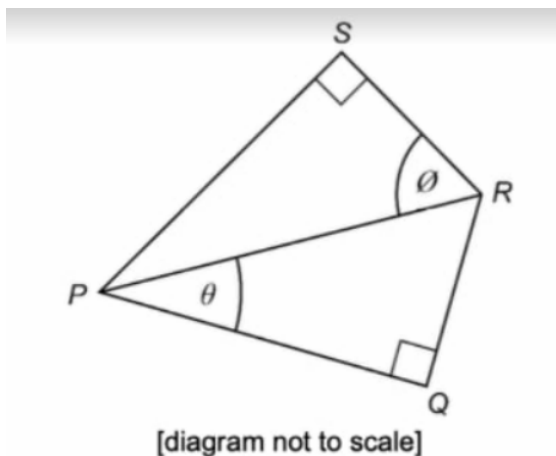
What is a percentage decrease in the area caused by the transformation

- a) 30%
- b) 35%
- c) 65%
- d) 70%

21) Two right-angled triangles are joined together to make a quadrilateral PQRS.

$$\tan(\theta) = \frac{3}{4}, \sin(\theta) = \frac{7}{9}.$$

Find the ratio of length PQ : length PS



a) 28:45

b) 20:63

c) 36:35

d) 4:7

22) When $x = 2$ is substituted in the expression $x^3 + px^2 + qx + p^2$, the result is 0.

When $x = 1$ is substituted into the same expression, the result is -3.5

Find all possible values of p

a) $p=1$

b) $p=1 \pm \frac{\sqrt{6}}{3}$

c) $p=1 \pm \sqrt{7}$

d) $p=-3$

23) Find a sum and write it as a fraction:

$0.(063) + 0.(64)$

a) $\frac{2353}{3669}$

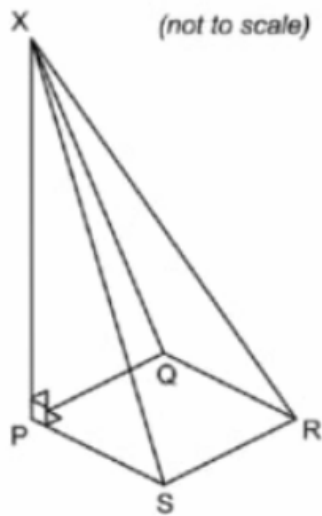
b) $\frac{2599}{3663}$

c) $\frac{2773}{3799}$

d) $\frac{2927}{3849}$

24) The diagram shows part of a glass structure. PQRS is a horizontal square with sides of 1 metre and point X is 4 meters vertically above P.

What is cosine of the angle that XR makes with the horizontal



a) $\frac{1}{3}$

b) $\frac{\sqrt{2}}{4}$

c) $\frac{\sqrt{2}}{2\sqrt{3}}$

d) $\frac{2}{3}$

25) Find the value of x, if:

$$5^x * 125^y = 3125^{23}$$

$$3^y + 3^y + 3^y = 243$$

- a) 93
- b) 103
- c) 113
- d) 123

26) Find the sum of the all solutions of the given inequality

$$|x - 5| < 6$$

- a) 39
- b) 49
- c) 55
- d) 66

27) The function f is given by $f(x) = x^3 + px^2 + qx - 8$ where p and q are real constants.

When $f(x)$ is divided by $(x + 1)$, the remainder is -27

When $f(x)$ is divided by $(x - 1)$, the remainder is -1

Find the reminder when $f(x)$ is divided by $(2x - 1)$

- a) $-\frac{25}{6}$

b) $-\frac{27}{8}$

c) $\frac{29}{9}$

d) $\frac{31}{11}$

28) $x^2 + (3m + 4)x + m - 1 = 0$ is given. If $x_1^2 + x_2^2 = 10$.

Find the one possible value of m

a) $1/2$

b) -2

c) $4/9$

d) $-2/9$

29) One alloy is 25% gold and the other is 40% gold.

How much of each should be used to produce 120 kg of an alloy that is 30% gold

a) 100kg(I) and 20kg(II)

b) 90kg(I) and 30kg(II)

c) 80kg(I) and 40kg(II)

d) 70kg(I) and 50kg(II)

30) If x, y, z are positive numbers, with $xy = 24$, $xz = 48$, $yz = 72$. Find $x + y + z$

a) 20

b) 22

с)28

d)36

как ты?

устал да?

я тоже

приляг отдохни

